

CrediFlow — Complete Project Master Briefing

AI Agent & Chatbot Reference Specification | End-to-End System Context

SYSTEM MISSION: CrediFlow is an enterprise fintech and statutory compliance engine that resolves **GST Input Tax Credit (ITC)** leakage for Indian businesses under **Rule 60 CGST** and **Section 16(2)(aa)**. It ingests multi-format Purchase Registers (PR) and GST Portal returns (GSTR-2B), runs sub-millisecond deterministic matching, coordinates an adversarial Dual-Agent AI audit system, safeguards high-value exposure with a Human-in-the-Loop (HITL) gate, and executes closed-loop recovery via bilingual nudges and court-ready legal PDF notices.

1. Statutory Context & The Problem CrediFlow Solves

- The GST ITC Bottleneck:** In India, a buyer pays GST to a supplier upon receiving goods/services. However, under Section 16(2)(aa) CGST Act, the buyer can *only* claim this Input Tax Credit if the supplier actually uploads the invoice to the government portal (GSTR-1), making it appear in the buyer's auto-generated GSTR-2B.
- Rule 60 CGST Strictness:** Strict zero-mismatch rule. If a supplier delays filing, reports the wrong buyer GSTIN, enters mismatched invoice numbers, misstates taxable values/rates, or defaults on their GSTR-3B tax payment, the buyer's ITC is frozen or clawed back with 18% annual statutory interest and penalties.
- Traditional Failure Mode:** Enterprise finance teams manually audit massive spreadsheets months late, causing massive working capital blockage, legal disputes, and irrecoverable tax write-offs.

2. CrediFlow Core System Capabilities

- Deterministic Math Engine:** 100% Python-based mathematical reconciliation (zero LLM hallucinations for tax amounts, CGST, SGST, IGST, and cess). Matches thousands of records in under 2 seconds.
- Adversarial Dual-Agent AI Audit:** Agent A (Root-Cause Classifier) diagnoses the statutory failure; Agent B (Audit Cross-Examiner) challenges the diagnosis against legal defenses, threshold materiality, and exemptions.
- Human-in-the-Loop (HITL) Gate:** Auto-routes transactions with high exposure (> INR 50,000) or low AI confidence (< 85%) to finance managers for approval, modification, or rejection before any vendor communication is sent.
- Closed-Loop Vendor Resolution:** Dispatches bilingual (English & Hindi) nudges via WhatsApp & Email, and dynamically compiles formal, court-ready Section 16(2)(aa) legal demand notices in PDF format.

3. Full Technology Stack Specification

Subsystem	Technologies & Libraries	Key Role in CrediFlow
Frontend UI	React 19.2, Vite 8.2, Tailwind CSS, Lucide React, Canvas Confetti, Oxlint	Interactive SPA dashboard, live ledger, HITL review gate, live AI feed, benchmark runner.
Backend API	FastAPI (0.115+), Uvicorn (ASGI), Pydantic v2, HTTPX, Python-Multipart, OpenPyXL	High-speed async REST endpoints, Excel/CSV/JSON file parsers, schema validation, batching.
Database	SQLite 3 (Local persistent / Vercel fallback) + FastAPI In-Memory Session Cache	Stores audit_ledger, benchmark_runs, human_decisions, and notice_dispatches.
Data Pipes	Tinybird / ClickHouse SQL Pipes (.pipe declarative format)	Streaming analytics, high-volume ingestion pipelines, and aggregate compliance reporting.
AI Engine	RocketRide Dual-Agent Pipeline (OpenAI backend) + Deterministic Python Fallback	Agent A Root Cause + Agent B Cross Examiner + 100% offline statutory fallback.
Doc Engine	ReportLab 5.0.1 + Custom Bilingual Nudge Generator	Generates formal legal demand PDF notices and bilingual English/Hindi notification drafts.
DevOps	Vercel Serverless (Python runtime + static Vite bundle), Pytest test suite	Production serverless deployment, automated compliance test suite under Rule 60.

4. End-to-End System Data Flow & Architecture

DATA FLOW LIFECYCLE:

Step 1 (Ingestion): User uploads Purchase Register (PR) & GSTR-2B (Excel/CSV/JSON) or generates demo batch.

Step 2 (Deterministic Match): GSTReconciliationEngine compares invoices, detects discrepancies, calculates exact ITC exposure.

Step 3 (Dual-Agent Audit): Discrepancies sent to RocketRide Webhook / Statutory Engine: Agent A classifies root-cause, Agent B cross-examines.

Step 4 (HITL Gate): High-risk (exposure > INR 50,000 or confidence < 85%) routed to Human Gate for review.

Step 5 (Resolution & Dispatch): Approved actions generate bilingual WhatsApp/Email nudges & ReportLab legal recovery PDFs.

Step 6 (Closed-Loop Update): Vendor filing simulation updates ledger status, logs to SQLite DB, and settles ITC.

5. Codebase Directory Structure & Key Files

File / Directory Path	Module Role & Core Responsibilities
backend/app/core/gst_reconciliation.py	Core deterministic rule engine implementing Rule 60 CGST. Classifies MATCH, VALUE_MISMATCH, RATE_MISMATCH, MISSING_IN_GSTR2B, etc.
backend/app/services/rocketride_service.py	Dual-agent AI orchestration. Priority: 1. RocketRide Webhook (OpenAI) -> 2. RocketRide SDK (WSS) -> 3. Deterministic Python Fallback.
backend/app/services/human_gate.py	Human-in-the-Loop approval gate. Manages queue, exposure thresholds, risk scoring, approvals, edits, and rejections.
backend/app/services/notice_generator.py	ReportLab PDF generator for formal Section 16(2)(aa) legal demand notices + bilingual Hindi/English nudge template engine.
backend/app/services/nudge_dispatcher.py	Simulated / active multi-channel communication gateway for WhatsApp Business API and Email SMTP delivery.
backend/app/core/db.py	SQLite 3 schema initialization and query operations for audit logs, benchmarks, human decisions, and notice dispatches.
backend/app/core/file_parser.py	Robust parser handling Excel (.xlsx), CSV, JSON files with automatic header mapping and normalization.
backend/app/api/routes.py	FastAPI REST router exposing all audit, benchmark, HITL decision, dispatch, and demo dataset endpoints.
frontend/src/App.jsx	Primary React single-page application orchestrating tabs, ledger tables, HITL modals, AI feeds, and live metrics.
pipelines/crediflow_audit_pipeline.pipe	ClickHouse / Tinybird streaming analytics definitions for high-volume invoice processing and real-time aggregation.

6. SQLite Database Schemas (data/crediflow.db)

- audit_ledger:** id, invoice_number, supplier_name, supplier_gstin, itc_exposure_rupees, mismatch_type, execution_engine, root_cause_code, audit_verdict, requires_human_review, created_at
- benchmark_runs:** id, run_id, count, runtime_ms, throughput_per_sec, total_exposure_inr, discrepancies_count, cost_usd, cost_inr, retries, escalations, created_at
- human_decisions:** id, gate_id (UNIQUE), invoice_number, decision (APPROVED|EDITED|REJECTED), decided_by, note, decided_at
- notice_dispatches:** id, invoice_number, supplier_name, channel (WHATSAPP|EMAIL), pdf_path, status, dispatched_at

7. Primary REST API Endpoints

- POST /api/upload: Uploads PR and GSTR-2B files (Excel/CSV/JSON) and runs instant deterministic matching.
- POST /api/audit: Triggers the RocketRide Dual-Agent AI audit on selected or all identified discrepancies.
- GET /api/gate/pending & POST /api/gate/decision: Manages the Human-in-the-Loop review queue and logs decisions.
- POST /api/dispatch/nudge: Dispatches bilingual nudges & compiles legal recovery PDF notices.
- GET /api/benchmark/run: Executes high-volume stress testing (100 to 10,000+ invoices) and returns performance metrics.
- GET /api/db/summary: Returns SQLite table counts, verification status, file size, and last benchmark.